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$$P_1. F = \int \frac{Gm dm}{(x-l)^2} = \int_0^L \frac{Gm}{(x-l)^2} dl \frac{M}{L} = \int_0^L \frac{GMm}{L} \frac{1}{(x-l)^2} dl = \frac{GMm}{x(L-L)}$$

when $x \gg L$ $F = \frac{GMm}{x^2}$

$$P_2. (a) U = \int \frac{Gm dm}{\sqrt{a^2+x^2}} = - \int_0^{2\pi a} \frac{Gm}{\sqrt{a^2+x^2}} \frac{M dl}{2\pi a} = - \frac{GMm}{\sqrt{a^2+x^2}} \quad C=0$$

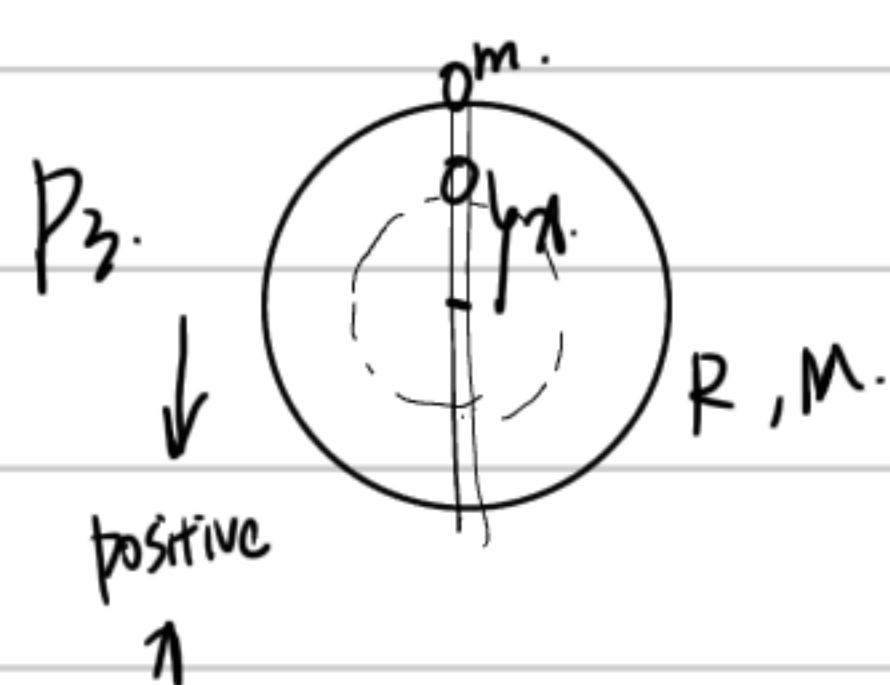
(b) $x \gg a$ $U = - \frac{GMm}{x}$ which is reasonable.

(c) $\tan \theta = \frac{a}{x}$ $\cos \theta = \frac{x}{\sqrt{a^2+x^2}}$
 $F = \int \frac{Gm dm}{a^2+x^2} \cos \theta = \int_0^{2\pi a} \frac{Gm}{a^2+x^2} \frac{M dl}{2\pi a} \cos \theta = \frac{GMm}{a^2+x^2} \frac{x}{\sqrt{a^2+x^2}}$ direction: towards left $\leftarrow m$

when $x \gg a$ $F = \frac{GMm}{x^2}$

(d) $U(x=0) = - \frac{GMm}{a}$ $F(x=0) = 0$

take positive x direction



$\rho = \frac{M}{\frac{4}{3}\pi R^3}$ for a ball with radius x , $M' = \frac{4}{3}\pi x^3 \rho$

$$F = - \frac{GM'm}{x^2} = - \frac{Gm}{x^2} \frac{4}{3}\pi x^3 \rho = - \frac{4}{3}\pi \rho Gm x$$

$\ddot{x} = a = - \frac{4}{3}\pi \rho G x$ so $\ddot{x} + \frac{4}{3}\pi \rho G x = 0$

it describes a harmonic oscillator

$$\omega^2 = \frac{4}{3}\pi \rho G \quad \omega = \sqrt{\frac{4}{3}\pi \rho G} \quad T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{4}{3}\pi \rho G}} = \frac{\sqrt{3}\pi}{\sqrt{\pi \rho G}} = \frac{2\pi \sqrt{R^3}}{\sqrt{GM}}$$



$T_{\text{sate}} = 2\pi \sqrt{\frac{R^3}{GM}} = T$

if the tunnel is drilled at an angle

$$F = - \frac{GM'm}{d^2+x^2} \cdot \cos \theta = - \frac{GM'm}{d^2+x^2} \frac{x}{\sqrt{d^2+x^2}} = - \frac{GMx}{(d^2+x^2)^{3/2}} \frac{4}{3}\pi \rho (d^2+x^2)^{3/2}$$

$\ddot{x} = a = - \frac{4}{3}\pi \rho G x$ so it's the same as the case above.

$T_{\text{angle}} = T_{\text{center}} = T_{\text{satellite}} = \frac{2\pi \sqrt{R^3}}{\sqrt{GM}}$